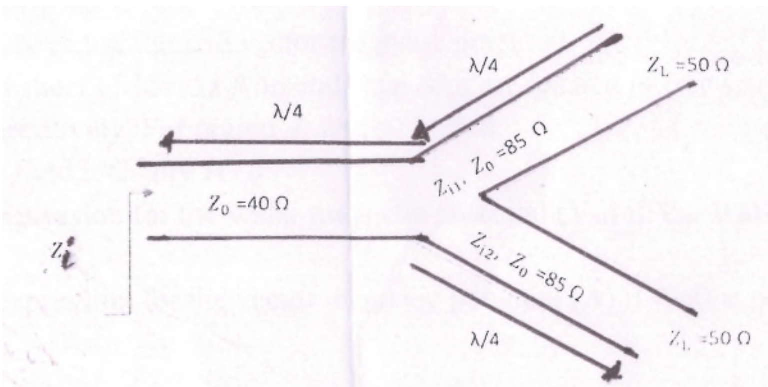
USN

RV COLLEGE OF ENGINEERING®
(An Autonomous Institution Affiliated to VTU)
IV Semester B. E. Examinations Oct-2023
Electronics and Communication Engineering
ELECTROMAGNETIC FIELDS AND APPLICATIONS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

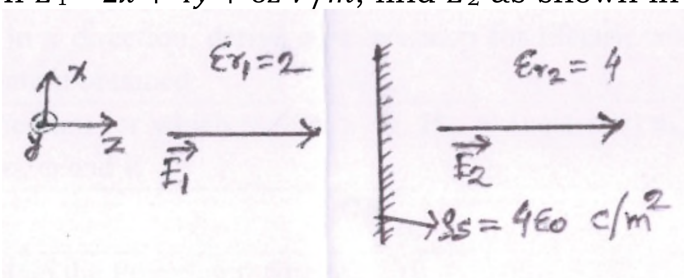
PART-A

1	1.1	Determine the divergence of the vector fields: i) $P = x^2yz\, a_x + xz\, a_z$ ii) $Q = \rho sin\varphi\, a_p + \rho^2 z\ a_\varphi + zcos\varphi\, a_z$	02
	1.2	A uniform line charge of $16nC/m$ is located along the line defined by $y = -2, z = 5m$. Find E at $P\ (1,2,3)\ m$.	02
	1.3	For a current distribution in free space $A = (2x^2y + yz)i + (xy^2 - xz^3)j - (6xyz - 2x^2y^2)k\ Wb/m$ Calculate the magnetic flux for $x = 1, 0 < y, z < 2$.	02
	1.4	A conductor of radius 'a' carries uniform current with $J = 20\, a_z A/m^2$. Find the vector magnetic potential for $\rho < a$.	02
	1.5	A rectangular coil with 100 windings and a length of $10\,cm$ and a width $6\,cm$ is initially held so that its plane is parallel to a $1.2\,T$ magnetic field. The loop is then rotated in $0.25s$ so that it is perpendicular to the magnetic field. Determine the induced emf in the loop.	02
	1.6	Let A=15 $sin\,(10^8t - 5x + 45°)a_z$. Is it a time harmonic vector field? If so, explain why. Find the phasor form of A.	02
	1.7	A $28m$ long transmission line with $Z_0 = 48\Omega$ operating at $100KHz$ is transmitted with a load $Z_L = 42 + 28j\Omega$. Determine the reflection coefficient Γ and $SWR(s)$.	02
	1.8	A transmission line terminates in two branches, each of length $\lambda/4$, as shown in Fig 1.8. The branches are terminated by 40Ω loads. The lines are lossless and have the characteristics impedances shown. Determine the impedance Z_i as seen by the source as given in the Fig 1.8.  Fig 1.8	02

	1.9	In case of good conductor, comment on the relationship between the attenuation constant and skin depth for a plane wave with the help of a neat diagram.	02
	1.10	In free space $(z \leq 0)$, a plane wave with $H = 20\, sin\,(0.5x\,10^8 t - \beta z) a_x\, mA/m$ is incident normally on a lossless medium $(\varepsilon = 4\varepsilon_0, \mu = 16\mu_0)$ in the region $z \geq 0$. Determine the reflection coefficient Γ and incident electric field component E_i.	02

PART-B

2	a	State Coulombs law in the context of electrostatics. Under what conditions it is valid. Point charges 1mC and $-2mC$ are located at $(3, 2, -1)$ and $(-1, -1, 4)$ respectively. Calculate the electric force on a 10nC charge located at $(0, 3, 1)$ and the electric field intensity at that point.	08
	b	Assume a solid conducting cylinder of radius R. Find the electric field intensity (E) at a point r inside $(r < R)$ and outside $(r > R)$ the cylinder. At which point the electric field intensity will be maximum? Discuss the same analysis for solid non-conducting cylinder. How they are different? Illustrate using a neat diagram of E versus r.	08
3	a	Explain Biot-Savart Law in the context of magnetics. Comment on the magnetic field for the following two cases i) If the point P is placed along 'a' finite straight current carrying wire ii) If the point P is placed at a distance 'a' for semi-infinite current carrying wire.	08
	b	For a long coaxial transmission line. The radius of inner conductor is 'a' and carries current I in $+z$ direction. The inner radius of outer conductor is 'b' and has a thickness of 't' and carries current I is carried in $-z$ direction. Calculate the magnetic field intensity (H) at i) $\rho \leq a$ ii) $a \leq \rho \leq b$ iii) $b \leq \rho \leq (b + t)$ iv) $\rho > b$ Illustrate the analysis for the above mentioned conditions with the help of a neat diagram of H versus ρ.	08
		OR	
4	a	Mention the properties of a solenoid. Derive an expression to find magnetic field (B) for the following cases of solenoid. i) At the end of solenoid ii) At the center of solenoid.	08
	b	Differentiate between scalar and vector magnetic potential. A plane current sheet of $K = 15z\ A/m$ and $-15z\ A/m$ are located in free space at $x = 0.2$ and $x = -0.2$, respectively. For region $-0.2 < x < 0.2$, find i) The magnetic field intensity H ii) Obtain the expression for the scalar magnetic potential (V_m) if V_m=0 at point $P\ (0.1, 0.2, 0.3)$. iii) Obtain the expression for the vector magnetic potential (A) if $A = 0$ at point $P\ (0.1, 0.2, 0.3)$.	08

5	a	State Lenz's law. Proof Maxwell's third equation in this context. Write in differential as well as in integral form. Distinguish between transformer emf and motional emf and prove that $\epsilon_m = \phi_l(\vec{v}x\vec{B}).\vec{dl}$, where ϵ_m is the emf generated due to motional loop. $\vec{v}$ is the velocity of the moving loop.	08
	b	State and proof the equation of continuity. For time varying fields, state and proof the modified form of Ampere's circuital law. Mention the same in both differential and integral form.	08
		OR	
6	a	Illustrate the boundary conditions $(z = 0)$ for the below mentioned cases, where dielectric medium $1 (z < 0)$ and dielectric medium $2 (z > 0)$ have conductivity σ_1 , permeability μ_1, permittivity ε_1 and conductivity σ_2 , permeability μ_2, permittivity ε_2 respectively. Assume surface charge component as ρ_s and current density K_s i) Normal and tangential components for electric flux density (D) and magnetic field (B) ii) Normal and tangential components for electric field (E) and magnetic field intensity (H) Also illustrate the change in analysis for the above mentioned components for dielectric-conductor medium and $\rho_s = 0$ and $K_s = 0$.	08
	b	i) If $E_1 = 2x + 4y + 6z\ V/m$, find E_2 as shown in Fig 6b.  Fig 6b ii) Let $\vec{B_s} = \frac{20}{j}\hat{a}_x + 10e^{j(\frac{2\pi x}{3})}\widehat{a_y}$. Find $\vec{B}$ as a function of space and time.	08
7	a	An open wire transmission line having primary constants $R\ ohm/m$, $L\ H/m$, $G\ mho/m$ and $C\ F/m$ operating at f Hz then i) Derive an expression for transmission line using distributed parameter model. ii) Bring out the similarities between Transmission Line Equation and Helmholtz's wave equation. If $V = V_s$ and $I = I_s$ at generator end Drive an expression to determine Voltage and current at any point l along the line.	08
	b	i) State the conditions for lossless transmission line. Derive the primary and secondary constants of the lossless transmission line. ii) A lossless transmission line of electrical length $l = 0.3\lambda$ is terminated with a complex load impedance as shown in Fig 7b.	

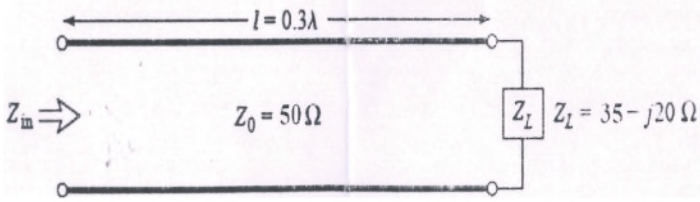


Fig 7b

		Find i) The reflection coefficient at the load ii) The SWR on the line.	08
		OR	
8	a	i) "When we have matched condition, there is no reflected wave" Comment on the validity of the statement. ii) Comment on the reflection coefficient if the line is short circuited, open circuited and terminated at ideal reactance. iii) The characteristics impedance of a transmission line is 75Ω. The input impedance of the open circuited line Z_{oc} is $50 + j70\Omega$. When the transmission line is short circuited, determine the value of the input impedance.	08
	b	i) "In case of transmission line, the maximum voltage and maximum current are staggered in space" Comment on the validity of the statement. ii) A $30m$ long lossless transmission line with characteristic impedance Z_o= 50Ω operating at $2MHz$ is terminated with a load Z_L= $60 + j40\ \Omega$. If $u = 0.6c$ on the line. Using Smith Chart i) The reflection coefficient at the load ii) The SWR on the line iii) The input impedance to the line.	08
9	a	i) Using Maxwell's equations derive Helmholt'z wave equation in lossy dielectric medium. ii) A Plane wave is propagating in lossy dielectric medium along z axis with electric field polarized in x direction, derive an expression for Electric and Magnetic wave using wave equation obtained.	08
	b	In a loss-less medium for which $\eta = 60\pi$, $\mu_r = 1$, $H = -0.1\, cos(\omega t - z)\, a_x + 0.5\, sin(\omega t - z)\, a_y\ A/m$, calculate ε_r, $\omega$and E.	08
		OR	
10	a	i) State and explain the Poynting theorem ii) In a non-magnetic medium $E = 4\, sin\,(2\pi \times 10^7 t - 0.8x) a_z\ V/m$, Find i) The time average power carried by the wave ii) The total power crossing 100 $sq - cm$ of the plane $2x + y = 5$.	08
	b	In free space ($z <= 0$), a plane wave with $H = 10cos(10^8 t - \beta z) a_x\ mA/m$ is incident normally on a lossless medium ($\varepsilon = 2\varepsilon_0, \mu = 8\mu_0$) in the region $z >= 0$. Determine the reflected wave H_r, E_r and the transmitted wave H_t, E_t.	08